

[Dietary Supplements for Exercise and Athletic Performance: Fact Sheet for Consumers \(original version\)](#)

Dietary Supplements for Exercise and Athletic Performance: Fact Sheet for Consumers

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What are dietary supplements for exercise and athletic performance and what do they do?

If you get regular exercise—and especially if you're an athlete and compete in sporting events—you know that a nutritionally adequate diet and plenty of fluids are important for maximizing your physical performance. You may wonder, however, if dietary supplements could help you train harder, improve performance, or gain a competitive edge.

This fact sheet describes what's known about the effectiveness and safety of many ingredients in dietary supplements that are promoted to improve exercise and athletic performance. These products are sometimes called ergogenic aids, but this fact sheet simply refers to them as performance supplements. Sellers of these supplements might claim that their products improve strength or endurance, help you achieve a performance goal more quickly, or increase your tolerance for more intense training. They might also claim that their supplements can help prepare your body for exercise, reduce the chance of injury during training, or assist with recovery after exercise.

Performance supplements cannot substitute for a healthy diet, but some of them may have value, depending on the type and intensity of your activity. Other supplements don't seem to work, and a few might be harmful.

If you're thinking about taking a performance supplement, talk to your health care provider. If you have a trainer or coach with knowledge of sports medicine, ask them about performance supplements. Talking to an expert is important if you're a teenager or have any medical conditions. It's also important to find out whether medications you take might interact with the performance supplements you're considering.

What are ingredients in supplements for exercise and athletic performance?

Performance supplements can contain many ingredients—like vitamins and minerals, protein, amino acids, and herbs—in different amounts and in many combinations. These products are sold in various forms, such as capsules, tablets, liquids, and powders.

This fact sheet describes ingredients in performance supplements below in alphabetical order. You'll learn whether each ingredient is effective and safe and get expert advice about using it. Keep in mind, however, that many performance supplements in the marketplace contain more than one ingredient, and ingredients can

work differently when they're combined. Because most ingredient combinations have not been studied, we don't know how effective or safe they are in improving performance.

You may be surprised to learn that makers of performance supplements usually don't carry out studies in people to find out whether their products really work and are safe. When studies on performance supplement ingredients and ingredient combinations are done (mainly by researchers at colleges and universities), they often involve small numbers of people taking the supplement for just a few days, weeks, or months. Much of the research is done in young healthy men, but not women, middle-aged and older adults, or teenagers. And often, studies haven't looked at the use of supplement ingredients or combinations in people involved in the same athletic activity as you. For example, the results from a study in weightlifters might not apply to you if you are a distance runner.

Ingredients in supplements for exercise and athletic performance

INGREDIENT	DOES IT WORK?	IS IT SAFE?	BOTTOM LINE
Antioxidants (vitamin C, vitamin E, and coenzyme Q10) You breathe in more oxygen when you exercise. As a result, free radicals form and damage muscle cells. Because antioxidants can reduce free-radical damage to muscle, some people think that taking them in a supplement might reduce muscle inflammation, soreness, and fatigue.	Does it work? No. The free radicals that form when you exercise seem to help muscle fibers grow and produce more energy. Antioxidant supplements might actually reduce some of the benefits of exercise, including muscle growth and power output. Also, they have little effect on aerobic fitness and performance in endurance activities like distance running.	Is it safe? Everyone needs adequate amounts of vitamin C and vitamin E for good health. Getting too much of these nutrients can be harmful, but the amounts of vitamin C (about 1,000 milligrams) and vitamin E (about 500 International Units [IU]) typically used in studies of performance supplements are below safe upper limits. The side effects from coenzyme Q10 can include tiredness, insomnia, headaches, and some gastrointestinal (GI) discomfort, but these effects tend to be mild.	Bottom Line There's little scientific evidence to support taking supplements containing vitamins C and E or coenzyme Q10 to improve performance if you're getting adequate amounts of these nutrients from a nutritious diet.
Arginine Arginine is an amino acid in foods that contain protein, like meat, poultry, fish, eggs, dairy products, and legumes. A nutritious diet supplies about 4 to 5 grams a day. Supplement sellers claim that taking larger amounts of arginine in supplements improves performance, partly because the body converts it into nitric oxide, which expands blood vessels and increases blood flow. Increased	Does it work? Although the research is limited, arginine supplements seem to have little to no effect on strengthening and muscle-building exercises (like bodybuilding) or aerobic activities (like running and cycling). Studies have used 2 to 20 grams a day of	Is it safe? Arginine supplements seem safe when users take up to 9 grams a day for several days or weeks. Taking more can cause GI discomfort and can slightly lower blood pressure.	Bottom Line There's little scientific evidence to support taking arginine supplements to increase strength, improve performance, or help tired and sore muscles recover after exercise.

Bottom Line: There's little scientific evidence to support taking supplements containing vitamins C and E or coenzyme Q10 to improve performance if you're getting adequate amounts of these nutrients from a nutritious diet.

blood flow helps deliver oxygen and nutrients to exercising muscle and speeds up the removal of waste products that cause muscle fatigue.

arginine for up to 3 months.

Bottom Line: There's little scientific evidence to support taking arginine supplements to increase strength, improve performance, or help tired and sore muscles recover after exercise.

Beetroot or beet juice

Beets and beet juice are among the best food sources of nitrate. Beet juice might improve athletic performance because the body converts some of this nitrate to nitric oxide, which expands blood vessels. This blood vessel expansion increases blood flow and the delivery of oxygen and nutrients to exercising muscle. The expanded blood vessels also speed up the removal of waste products that cause muscle fatigue.

Does it work?

Many, but not all, studies have found that beet juice can improve performance and endurance in aerobic activities like running, swimming, cycling, and rowing, but whether it helps with strengthening and bodybuilding exercises isn't known. Beet juice is more likely to improve the performance of recreational exercisers than highly trained athletes. The usual approach in studies is for participants to drink 2 cups of beet juice about 2.5 to 3 hours before exercise.

Is it safe?

Drinking moderate amounts of beet juice is safe, but it can turn your urine pink or red.

Bottom Line

Beet juice might improve aerobic exercise performance if you're recreationally active. But whether dietary supplements containing beetroot powder have the same effects as beet juice isn't known.

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Beta-alanine

Beta-alanine is an amino acid in foods such as meat, poultry, and fish. People get up to about 1 gram a day of beta-alanine, depending on their diet. Your body uses beta-alanine to make carnosine in skeletal muscles. When you exercise intensely for several minutes, your muscles produce lactic acid, which reduces muscular force and causes tiredness. Carnosine reduces the buildup of lactic acid. Beta-alanine supplements increase muscle carnosine levels by different amounts, depending on the person.

Does it work?

Some, but not all, studies have shown that beta-alanine produces small performance improvements in swimming and team sports, like hockey and football, that require high-intensity, intermittent effort over short periods. Whether beta-alanine helps with endurance activities like cycling isn't clear. It's also not clear whether beta-alanine mainly benefits trained athletes or recreational exercisers. In most studies, participants took 1.6 to 6.4 grams a day of beta-alanine for 4 to 8 weeks.

Is it safe?

Taking 800 milligrams or more beta-alanine can cause moderate to severe paresthesia, a tingling, prickling, or burning sensation in your face, neck, back of the hands, and upper trunk. This effect can last 60 to 90 minutes but is not considered serious or harmful. Taking divided doses or a sustained-release form of beta-alanine can reduce or eliminate this paresthesia. It isn't known whether it's safe to take beta-alanine supplements daily for more than several months.

Bottom Line

Sports-medicine experts disagree on the value of taking beta-alanine supplements to enhance performance in high-intensity, intermittent activities. The International Society of Sports Nutrition recommends that if you are healthy and want to try beta-alanine supplements, take a daily loading dose of 4 to 6 grams per day (in divided doses with meals) for at least 2 weeks to see if it helps.

Bottom Line: Sports-medicine experts disagree on the value of taking beta-alanine supplements to enhance performance in high-intensity, intermittent activities. The International Society of Sports Nutrition recommends that if you are healthy and want to try beta-alanine supplements, take a daily loading dose of 4 to 6 grams per day (in divided doses with meals) for at least 2 weeks to see if it helps.

Beta-hydroxy-beta-methylbutyrate (HMB) Your body converts a small amount of leucine, one of the amino acids in foods and protein powders, to HMB. Your liver then converts the HMB into another compound that experts think helps muscle cells restore their structure and function after exercise. HMB also helps build protein in muscle and reduces muscle-protein breakdown.	Does it work? It's hard to know whether you might benefit from using HMB supplements because the research on these supplements has included adults of very different ages and fitness levels who took widely varying doses for different amounts of time. Overall, HMB seems to speed up recovery from exercise that's intense enough and long enough to cause muscle damage. Therefore, if you're a trained athlete, you'll need to exert yourself more than recreationally active people to cause the muscle damage that HMB might help treat.	Is it safe? Studies haven't reported any side effects in adults taking 3 grams per day of HMB for up to 8 weeks.	Bottom Line It's not clear whether taking HMB supplements will improve athletic performance. The International Society of Sports Nutrition recommends that if you are a healthy adult who wants to try HMB supplements, you should take 3 grams per day in three equal servings of 1 gram for at least 2 weeks to see if it helps. HMB comes in two forms: one with calcium and one without. A dose of 3 grams of the type with calcium supplies about 400 milligrams of calcium.
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Bottom Line: It's not clear whether taking HMB supplements will improve athletic performance. The International Society of Sports Nutrition recommends that if you are a healthy adult who wants to try HMB supplements, you should take 3 grams per day in three equal servings of 1 gram for at least 2 weeks to see if it helps. HMB comes in two forms: one with calcium and one without. A dose of 3 grams of the type with calcium supplies about 400 milligrams of calcium.

Betaine Your body makes betaine, and it is also found in foods such as beets, spinach, and whole-grain bread. You get about 100 to 300 milligrams a day of betaine when you eat a nutritious diet. How betaine supplements might affect or improve your performance isn't known.	Does it work? Only a few, mostly small, studies have evaluated betaine as a performance supplement. Most of these studies examined the use of betaine supplements to improve strength and power performance in bodybuilders. The studies found either no performance improvements or only modest ones. Participants in these studies took 2 to 5 grams a day of betaine for up to 15 days.	Is it safe? The few studies in which athletes took betaine supplements didn't find any side effects, but there hasn't been enough research to know for sure whether it's really safe.	Bottom Line There's little scientific evidence to support taking betaine supplements to improve performance if you eat a nutritious diet.
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Branched-chain amino acids (BCAAs)

The amino acids leucine, isoleucine, and valine are known as BCAAs. Animal foods, like meat, fish, and milk, contain BCAAs. Your muscles can use these three amino acids to provide energy during exercise. Leucine might also help build muscle.

Does it work?

There's little evidence that BCAA supplements improve performance in endurance activities like distance running. BCAA supplements might help increase your muscle size and strength together with a weight-training program, but it isn't clear whether taking BCAA supplements will help you build more muscle than just eating enough high-quality protein foods.

Is it safe?

A nutritious diet with enough protein can easily provide 10 to 20 grams a day of the BCAAs. Taking up to another 20 grams a day of BCAAs in supplements seems to be safe.

Bottom Line

There's not much scientific evidence to support taking BCAA supplements to improve performance, build muscle, or help tired and sore muscles to recover after exercise. Eating foods containing protein automatically increases your intake of BCAAs.

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Caffeine

Caffeine is a stimulant in beverages (like coffee, tea, and energy drinks) and in herbs (such as guarana and kola nut). Caffeine is also added to some dietary supplements. Moderate amounts of caffeine might increase your energy levels and reduce fatigue for several hours.

Does it work?

Caffeine might improve endurance, strength, and power in team sports. It's most likely to help with endurance activities (such as distance running) and sports that require intense, intermittent effort (like soccer and tennis). Caffeine doesn't help with short, intense exercise like sprinting or weightlifting. People have different responses to caffeine. It doesn't boost performance in everyone, or it may only slightly boost performance.

Is it safe?

Caffeine intakes of up to 400 to 500 milligrams a day seem safe in adults. Teenagers should limit their caffeine intake to no more than 100 milligrams a day. Taking 500 milligrams or more a day can reduce rather than improve physical performance, disturb sleep, and cause irritability and anxiety. Taking 10,000 milligrams or more in a single dose (one tablespoon of pure caffeine powder) can be fatal.

Bottom Line

Sports-medicine experts agree that caffeine can help you exercise at the same intensity level for longer and reduce feelings of fatigue. They suggest taking 2 to 6 milligrams per kilogram of body weight 15 to 60 minutes before you exercise. The National Collegiate Athletic Association and International Olympic Committee limit the amount of caffeine that athletes can take before a competition.

The usual dose of caffeine to aid performance is 2 to 6 milligrams per kilogram of body weight, or about 210 to 420 mg caffeine for a 154-pound person. (By comparison, a cup of coffee has about 85 to 100 milligrams of

caffeine.) Taking more probably doesn't improve performance further and can increase the risk of side effects.

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Citrulline

Citrulline is an amino acid that your body produces; it is also present in some foods. Your kidneys convert most citrulline into another amino acid, arginine. Your body then transforms the arginine into nitric oxide, which expands blood vessels. This expansion increases blood flow and the delivery of oxygen and nutrients to exercising muscles and speeds up the removal of waste products that cause muscle fatigue.

Does it work?

The research on citrulline as a performance supplement is limited. A few studies find that citrulline might help improve, hinder, or have no effect on performance. In these studies, participants took up to 9 grams of citrulline for 1 day or 6 grams per day for up to 16 days.

Is it safe?

There isn't enough research on citrulline to know for sure whether it's safe. Some users have reported that it can cause stomach discomfort.

Bottom Line

There's not much scientific evidence to support taking citrulline supplements to improve exercise or athletic performance.

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Creatine

Creatine is a compound that is stored in your muscles and supplies them with energy. Your body produces some creatine (about 1 gram a day), and you get some creatine from eating animal-based foods, such as beef and salmon (about 500 milligrams in a 4-ounce serving). However, it is only when you take much larger amounts of creatine from dietary supplements that it might improve certain types of performance.

Does it work?

Creatine supplements can increase strength, power, and the ability to contract muscles for maximum effort, but the extent of performance improvements from creatine supplements differs among individuals.

Use of creatine supplements for several weeks or months can help with training. Overall, creatine enhances performance during repeated short bursts of intense, intermittent activity (lasting up to about 2.5 minutes at a time), such as sprinting and weightlifting. Creatine seems to have little value for endurance activities, such as distance running, cycling, or swimming.

Is it safe?

Creatine is safe for healthy adults to take for several weeks or months. It also seems safe for long-term use over several years. Creatine usually causes some weight gain because it increases water retention. Rare individual reactions to creatine include some muscle stiffness and cramps as well as GI distress.

Bottom Line

Sports-medicine experts agree that creatine supplements can improve performance in activities that involve intense effort followed by short recovery periods. It can also be valuable in training for certain athletic competitions. In studies, people often took a loading dose of about 20 grams per day of creatine (in four equal portions) for 5 to 7 days and then 3 to 5 grams a day. Creatine monohydrate is the most widely used and studied form of creatine in supplements.

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Deer antler velvet Deer antler velvet supplements are made from the antlers of deer or elk before the antlers turn into bone. Deer antlers might contain growth factors that could promote muscle growth.	Does it work? There's been little research on use of deer antler velvet to improve performance in either strength or endurance activities. The few published studies have found no benefit from taking the supplement.	Is it safe? Deer antler velvet hasn't been studied enough to know whether taking it is safe.	Bottom Line There's no scientific evidence to support taking deer antler velvet supplements to improve exercise or athletic performance.
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Bottom Line: There's no scientific evidence to support taking deer antler velvet supplements to improve exercise or athletic performance.

Dehydroepiandrosterone (DHEA) DHEA is a steroid hormone produced by the adrenal glands. Your body converts some DHEA into testosterone, the male hormone that enhances muscle size and strength.	Does it work? There's been little study of the use of DHEA supplements to improve performance. The few published studies (all in men) have found no benefit from taking the supplement. Muscle size or strength and aerobic capacity didn't improve, and testosterone levels didn't rise.	Is it safe? DHEA hasn't been studied enough to know whether it's safe to take. Two small studies in men found no side effects, but in women, taking DHEA supplements for months can increase testosterone levels, which can cause acne and facial hair growth.	Bottom Line There's no scientific evidence to support taking DHEA to improve exercise or athletic performance. The National Collegiate Athletic Association and the World Anti-Doping Agency prohibit the use of DHEA in athletic competitions.
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Bottom Line: There's no scientific evidence to support taking DHEA to improve exercise or athletic performance. The National Collegiate Athletic Association and the World Anti-Doping Agency prohibit the use of DHEA in athletic competitions.

Ginseng Ginseng is the root of a plant used for thousands of years in traditional Chinese medicine. Some experts believe that <i>Panax</i> (also known as Chinese, Korean, Japanese, or American) ginseng might improve stamina and vitality. Siberian or Russian ginseng has been used to fight fatigue and strengthen the immune system.	Does it work? Several small studies have examined whether <i>Panax</i> or Siberian ginseng supplements can improve performance. This research provides little evidence that various doses and preparations of these supplements improve performance in athletes or recreational exercisers.	Is it safe? Both <i>Panax</i> and Siberian ginseng seem to be safe. However, ginseng supplements can cause headaches or GI effects and disturb sleep.	Bottom Line There's little scientific evidence to support taking ginseng supplements to improve exercise or athletic performance.
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Bottom Line: There's little scientific evidence to support taking ginseng supplements to improve exercise or athletic performance.

Glutamine Glutamine is an amino acid that your body uses to produce energy. Adults consume about 3 to 6 grams a day from protein-containing	Does it work? Only a few studies have examined the use of glutamine supplements for improving	Is it safe? Studies haven't reported any side effects from the use of up to 45 grams a day	Bottom Line There's little scientific evidence to support taking glutamine supplements to
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foods such as meat, poultry, fish, eggs, dairy products, and legumes. Your body also makes some glutamine, mainly from [BCAAs](#).

performance in strengthening and muscle-building exercises (like bodybuilding) and for recovering from these exercises (for example, by reducing muscle soreness). Glutamine has either no effect or provides only a small benefit.

of glutamine for several weeks in adults.

improve exercise or athletic performance.

Bottom Line: There's little scientific evidence to support taking glutamine supplements to improve exercise or athletic performance.

Iron

Iron is a mineral that delivers oxygen to muscles and tissues throughout your body. Cells also need iron to turn food into energy. Iron deficiency, especially with anemia, limits your ability to exercise and be active because it makes you tired and reduces your performance.

The recommended amount of iron to get each day is 11 milligrams for teenage boys, 15 milligrams for teenage girls, 8 milligrams for men to age 50, 18 milligrams for women to age 50, and 8 milligrams for older adults of both sexes. Recommended amounts are even higher for athletes, vegetarians, and vegans. Teenage girls and premenopausal women have the greatest risk of not getting enough iron from their diets.

Bottom Line: Taking enough iron in supplements to treat iron-deficiency anemia improves exercise capacity, but a healthcare provider should diagnose this condition before you start taking iron supplements. If you want to improve your athletic performance, you should eat a healthy diet containing foods rich in iron, such as lean meats, seafood, poultry, beans, nuts, and raisins. If needed, an iron-containing dietary supplement can help you get the recommended amount of iron.

Protein

Protein helps to build, maintain, and repair your muscles. It improves your body's response to athletic training and helps shorten the time you need to recover after exercise. Protein is made from amino acids. Your body makes some amino acids but needs to get others (known as essential amino acids [EAAs]) from food. Animal foods like meat, poultry, fish, eggs, and dairy products contain all of the EAAs. Plant foods like grains

Does it work?

Adequate protein in your diet provides the EAAs necessary for making muscle proteins and reduces the breakdown of proteins in your muscles. Athletes need about 0.5 to 0.9 grams of protein per pound of body weight a day (or about 75 to 135 grams for a person weighing 150

Is it safe?

High intakes of protein seem to be quite safe, but there is no benefit to consuming more than recommended amounts.

Bottom Line

If you are an athlete, you can probably eat enough foods that contain protein to meet your needs for protein. If needed, protein supplements and protein-fortified food and beverage products can help you get enough protein. Sports-science experts recommend that

and legumes contain different EAAs, so eating a diet containing different types of plant-based foods is one way to get all EAAs. Most protein powders and drinks contain whey, a protein in milk that provides all the EAAs.

pounds). You might need even more for a short time when you're training intensely or if you reduce your food intake to improve your physique or achieve a competition weight.

athletes consume 0.14 grams of protein per pound of body weight (about 20 grams for a person weighing 150 pounds) of high-quality protein (from animal foods and/or a mix of different plant foods) every 3 to 5 hours, including before sleep and within 2 hours after exercising.

Bottom Line: If you are an athlete, you can probably eat enough foods that contain protein to meet your needs for protein. If needed, protein supplements and protein-fortified food and beverage products can help you get enough protein. Sports-science experts recommend that athletes consume 0.14 grams of protein per pound of body weight (about 20 grams for a person weighing 150 pounds) of high-quality protein (from animal foods and/or a mix of different plant foods) every 3 to 5 hours, including before sleep and within 2 hours after exercising.

Quercetin

Quercetin is a compound found in fruits, vegetables, and some beverages (like tea). Some experts suggest that quercetin supplements increase energy production in muscle and improve blood flow throughout your body. A nutritious diet provides up to about 13 milligrams a day of quercetin.

Does it work?

There's limited research on the use of quercetin supplements to improve performance. The studies found that any benefits, when they occur, tend to be small. In these studies, participants took about 1,000 milligrams a day of quercetin for up to 8 weeks.

Is it safe?

The studies of quercetin supplements didn't find any side effects in the athletes who took them, but quercetin hasn't been studied enough to know whether it's really safe.

Bottom Line

There's little scientific evidence to support taking quercetin supplements to improve exercise or athletic performance.

Bottom Line: There's little scientific evidence to support taking quercetin supplements to improve exercise or athletic performance.

Ribose

Ribose is a natural sugar your body makes that helps with energy production in muscle. Some scientists believe that ribose supplements help muscles produce more energy.

Does it work?

There's been little study of the use of ribose supplements to improve performance. The few published studies in both trained athletes and occasional exercisers have shown little if any benefit from doses ranging from 625 milligrams to 10,000 milligrams a day for up to 8 weeks.

Is it safe?

The studies of athletes taking ribose supplements have found no side effects, but ribose hasn't been studied enough to know whether it's really safe when taken in large amounts for several months or more.

Bottom Line

There's very little scientific evidence to support taking ribose supplements to improve exercise or athletic performance.

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Sodium bicarbonate

Sodium bicarbonate is commonly known as baking soda. Exercising intensely over several minutes causes muscles to produce acids,

Does it work?

Studies show that athletes who take sodium bicarbonate might improve their

Is it safe?

Sodium bicarbonate can cause GI distress, including nausea and vomiting, and weight

Bottom Line

Sodium bicarbonate might provide some performance benefit in strenuous exercise that

such as lactic acid, that reduce muscle force and cause tiredness. Sodium bicarbonate can reduce the buildup of these acids.	performance a little in intense, short-term activities (like sprinting and swimming) and in intermittently intense sports (like tennis and boxing). However, different athletes respond differently to sodium bicarbonate. Sodium bicarbonate might actually hinder performance in some people. The usual dose taken is 300 milligrams per kilogram of body weight, or about 4 to 5 teaspoons of baking soda. Some people find this amount of sodium bicarbonate, dissolved in liquid, too salty to drink.	gain due to water retention. It is also high in sodium (1,260 milligrams per teaspoon).	lasts several minutes and in sports that require intermittent, intense activity, especially for trained athletes. However, in some people, sodium bicarbonate provides no performance benefit, and it can even reduce performance.
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Bottom Line: Sodium bicarbonate might provide some performance benefit in strenuous exercise that lasts several minutes and in sports that require intermittent, intense activity, especially for trained athletes. However, in some people, sodium bicarbonate provides no performance benefit, and it can even reduce performance.

Tart or sour cherry Tart or sour cherries of the Montmorency variety contain compounds that might help you recover from strenuous exercise. Specifically, these cherries might help to reduce pain, muscle damage from strength-related activities, and lung trauma from endurance activities that require deep, heavy breathing.	Does it work? There's limited research on tart cherry as a performance supplement. The studies that have been done suggest that it might help bodybuilders recover their strength faster and feel less muscle soreness after exercising. The supplements could also help runners race faster and be less likely to develop a cold or respiratory problem after a marathon. The typical dose is about 2 cups of juice or 500 milligrams of tart-cherry-skin powder for a week before the exercise and for 2 days afterwards.	Is it safe? Studies of tart-cherry products in athletes have not found any side effects, but the safety of tart-cherry supplements has not been well studied.	Bottom Line There's limited scientific evidence to support taking tart-cherry products to improve exercise and athletic performance.
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Bottom Line: There's limited scientific evidence to support taking tart-cherry products to improve exercise and athletic performance.

Tribulus terrestris <i>Tribulus terrestris</i> is a plant	Does it work? There's limited research	Is it safe? <i>Tribulus terrestris</i>	Bottom Line There's no scientific
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containing compounds that some sellers claim can improve performance by increasing levels of several hormones, including the male hormone testosterone.	on the use of <i>Tribulus terrestris</i> supplements to increase strength or muscle mass. The few studies investigating it did not find that it had any benefit.	hasn't been studied enough to know whether it's safe. Studies in animals show that high doses can cause heart, liver, and kidney damage.	support for taking <i>Tribulus terrestris</i> supplements to improve exercise or athletic performance. Some sports-medicine experts advise against taking any dietary supplements claimed to boost testosterone.
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Bottom Line: There's no scientific support for taking *Tribulus terrestris* supplements to improve exercise or athletic performance. Some sports-medicine experts advise against taking any dietary supplements claimed to boost testosterone.

How does the U.S. government regulate dietary supplements for exercise and athletic performance?

The U.S. Food and Drug Administration (FDA) regulates dietary supplements for exercise and athletic performance differently from prescription or over-the-counter drugs. As with other dietary supplements, FDA does not test or approve performance supplements before they are sold. Manufacturers are responsible for making sure that their supplements are safe and that the claims on the product labels are truthful and not misleading.

When FDA finds an unsafe dietary supplement, it can remove the supplement from the market or ask the supplement maker to recall the product. FDA and the Federal Trade Commission can also take action against companies that make false performance-improvement claims about their supplements; add pharmaceutical drugs or other adulterants to their supplements; or claim that their supplements can diagnose, treat, cure, or prevent a disease.

For more information about dietary supplement regulations, see the Office of Dietary Supplements (ODS) publication, [Dietary Supplements: What You Need to Know](#).

Can dietary supplements for exercise and athletic performance be harmful?

Like all dietary supplements, performance supplements can have side effects and might interact with prescription and over-the-counter medications. Many of these products contain multiple ingredients that have not been adequately tested in combination with each another.

Interactions with medications

Some dietary supplements for improving exercise and athletic performance can interact or interfere with other medications or supplements. For example, ginseng can reduce the blood-thinning effects of warfarin (Coumadin). Cimetidine (Tagamet HB, used to treat duodenal ulcers) can slow the removal of caffeine from the body and thus increase the risk of side effects from caffeine consumption.

If you take dietary supplements and medications on a regular basis, tell your health care provider.

Fraudulent and adulterated products

FDA warns that some products marketed as dietary supplements to improve exercise and athletic performance might contain inappropriate, unlabeled, or unlawful stimulants, steroids, hormone-like ingredients, controlled substances, prescription medications, or unapproved drugs. Using these tainted products can cause health problems and disqualify athletes from competitions.

FDA prohibits certain ingredients that some performance dietary supplements used to contain. These prohibited ingredients include androstenedione, dimethylamylamine (DMAA), and ephedra. Not only are these ingredients unsafe, but there is no scientific evidence showing that they can improve performance.

Sellers of some performance supplements ask certain companies to evaluate their products and certify that they are free from many banned ingredients and drugs. The major companies providing this certification service are [NSF](#) through its Certified for Sport® program, [Informed-Choice](#), and the [Banned Substances Control Group](#). Products that pass these tests may carry the certifier's official logo and are listed on the certifier's website.

Choosing a sensible approach to improving exercise and athletic performance

If you are a competitive or recreational athlete, you will perform at your best and recover most quickly when you eat a nutritionally adequate diet, drink enough fluids, are physically fit, and are properly trained. Only a few dietary supplements have enough scientific evidence showing that they can improve certain types of exercise and athletic performance. Athletes might use these supplements, if interested, if they already eat a good diet, train properly, and obtain guidance from a health care provider or sports-medicine expert.

In most cases, only adults should use performance supplements. The American Academy of Pediatrics, for example, states that performance supplements don't improve the abilities of teenage athletes beyond those that come from proper nutrition and training.

Where can I find out more?

- For general information on dietary supplements to improve performance
 - ODS Health Professional Fact Sheet on [Dietary Supplements for Exercise and Athletic Performance](#)
- For more advice on choosing dietary supplements
 - ODS [Frequently Asked Questions: Which brand\(s\) of dietary supplements should I purchase?](#)
- For information about building a healthy dietary pattern
 - [Dietary Guidelines for Americans](#)
 - [MyPlate](#)

Glossary

adverse event

An unwanted medical problem that occurs during treatment. Adverse events may be unrelated to the treatment or they may be caused by the therapy or procedure. For example, an adverse event may be caused by the toxic effects of a particular drug or dietary supplement or by an interaction with another therapy. Also called adverse effect and side effect.

American Academy of Pediatrics

AAP. An organization of pediatricians (medical doctors who specialize in the development, care, and diseases of children) that works to improve the health and well-being of infants, children, adolescents, and young adults.

amino acid

A chemical building block of protein.

capsule

A gelatin shell containing a dose of medicine, a vitamin, or other dietary supplement.

control

In a clinical trial, the group of participants that does not receive the new treatment being studied. This group is compared with the group receiving the new treatment, to see whether the new treatment works. In an observational study, the controls are participants who do not have a particular health condition; the control group is compared with the group of participants who do have the condition to

see if certain factors (such as diet, activity level, or use of dietary supplements) may be associated with developing or preventing the condition.

cure

To heal or restore health; a treatment to restore health.

diagnose

The process of using signs and symptoms to identify a disease.

dietary supplement

A product that is intended to supplement the diet. A dietary supplement contains one or more dietary ingredients (including vitamins, minerals, herbs or other botanicals, amino acids, and other substances) or their components; is intended to be taken by mouth as a pill, capsule, tablet, or liquid; and is identified on the front label of the product as being a dietary supplement.

drug

Any substance (other than food) that is used to prevent, diagnose, treat, or relieve symptoms of a disease or abnormal condition. Also, a substance that alters mood or body function or that can be habit-forming or addictive, especially a narcotic.

effectiveness

In medicine, the ability of an intervention (for example, a drug, surgery, or a dietary supplement) to produce the desired beneficial effect under the usual conditions of care by a health care provider.

evidence

Information used to support the use of a particular screening procedure, treatment, or preventive measure. In medicine, evidence needed to determine effectiveness is provided by laboratory research, clinical trials, and other studies.

Federal Trade Commission

FTC. A federal agency that protects consumers by preventing deceptive and unfair business practices. This includes unfair or deceptive advertising and marketing practices.

Food and Drug Administration

FDA, Department of Health and Human Services. FDA is the Federal government agency responsible for ensuring that foods and dietary supplements are safe, wholesome and sanitary, and that drugs, medical devices, cosmetics, and food are honestly, accurately and informatively represented to the public. FDA regulates dietary supplements under a different set of regulations than those covering conventional foods and drug products (prescription and over-the-counter). The dietary supplement manufacturer is responsible for ensuring that a dietary supplement is safe before it is marketed. FDA is responsible for taking action against any unsafe dietary supplement product after it reaches the market. Generally, manufacturers do not need to get FDA approval before producing or selling dietary supplements.

herb

A plant used in cooking, in tea, and for medicinal purposes.

hormone

A group of chemicals made by glands in the body. Hormones circulate in the bloodstream and control the actions of certain cells or organs. Some hormones can also be manufactured.

ingredient

In a dietary supplement, an ingredient is a component of the product, such as the main nutrient (vitamin, mineral, herb, amino acid, or enzyme) or any binder, color, filler flavor, or sweetener. In herbal supplements, the common name and Latin name (the genus and species) of the plant is given in the ingredient list. On a dietary supplement label, the ingredients are listed by weight, with the ingredient used in the largest amount first on the list and the ingredient used in the least amount at the end of the list.

interaction

A change in the way a dietary supplement acts in the body when taken with certain other supplements, medicines, or foods, or when taken with certain medical conditions. Interactions may cause the dietary supplement to be more or less effective, or cause effects on the body that are not expected.

label

When referring to dietary supplements, information that appears on the product container, including a descriptive name of the product stating that it is a "supplement"; the name and place of business of the manufacturer, packer, or distributor; a complete list of ingredients; and each dietary ingredient contained in the product. Supplements must also include directions for use, nutrition labeling in the form of a Supplement Facts panel that identifies each dietary ingredient contained in the product and the serving size, amount, and active ingredients.

mineral

In nutrition, an inorganic substance found in the earth that is required to maintain health.

nutrition

The process of eating, digesting, and absorbing nutrients (such as protein, carbohydrate, fat, vitamins, minerals, and water) from food to maintain the body, grow new cells, repair tissues, and supply energy. Nutrition is also the science of food, diet, and health.

nutritional

Having to do with nutrition (eating, digesting, and absorbing the nutrients in food, and the health and disease consequences).

Office of Dietary Supplements

ODS, Office of Disease Prevention, Office of Director, National Institutes of Health, Department of Health and Human Services. ODS strengthens knowledge and understanding of dietary supplements by evaluating scientific information, stimulating and supporting research, disseminating research results, and educating the public to foster an enhanced quality of life and health for the US population.

prescription

A written order from a health care provider for medicine, therapy, or tests.

prevent

To stop from happening.

protein

A molecule made up of amino acids that the body needs for good health. Proteins are the basis of body structures such as skin and muscle, and substances such as enzymes and antibodies.

regulate

To govern, make uniform, and bring under the control of a rule, principle, or legal system. In the United States, the FDA has the authority to regulate dietary supplements.

risk

The chance or probability that a harmful event will occur. In health, for example, the chance that someone will develop a disease or condition.

stimulant

A substance that increases brain activity, alertness, attention, blood pressure, heart rate, breathing, and energy.

supplement

A nutrient that may be added to the diet to increase the intake of that nutrient. Sometimes used to mean dietary supplement.

tolerance

The ability to take a drug or dietary supplement without discomfort or unwanted side effects. Also, a condition that occurs when the body gets used to a drug or dietary supplement so that either larger amounts or a different drug or supplement is needed to get the same effect originally experienced.

treat

To care for a patient with a disease by using medicine, surgery, or other approaches.

ulcer

A sore on the skin or in the lining of an organ or other tissue that deepens as cells die. Ulcers form where swelling, redness, pain, and warmth (inflammation) have killed the cells, then bleed and produce pus.

vitamin

A nutrient that the body needs in small amounts to function and maintain health. Examples are vitamins A, C, and E.

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